



Bengaluru Road Accident Early Warning System

AI-Powered Predictive Safety for Namma City's Roads

MSTAT509A-4 Project | Christ (Deemed to be University)

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Shifting Traffic Management

From Reactive to Predictive

Current Reactive Approach

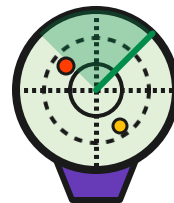
The current reactive approach relies on tabulating accident totals only **after the year ends**, resulting in **delayed police action** at crowded junctions like **Silk Board**.



Our AI Early Warning Solution

Our AI Early Warning Solution introduces **1-year advance forecasting**.

This enables **proactive police deployment** by identifying the **top-10 high-risk zones** before crashes happen, transforming road safety management.



AI for Safer Bengaluru Roads



Mission Brief: Spatial Predictors & Deployment

01. The Data

Clean 7-year accident database spanning 2018 to 2024.

Covers 64 police station jurisdictions across Bengaluru for deep spatial-temporal tracking.

02. Feature Engineering

Time-aware crash metrics & rolling velocity.

Engineered time-aware crash metrics and 2-year rolling velocity features to capture localized risk trends.

03. Predictive Model

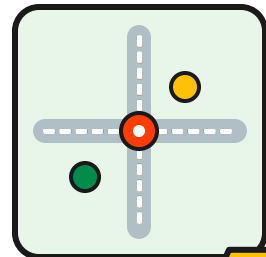
Class-weighted neural network targeting next-year hotspots.

Trained a class-weighted neural network targeting next-year top hotspots to optimize policing resource allocation.

04. Folium Bengaluru map Deployment

Interactive color-coded dashboard.

Interactive color-coded Folium Bengaluru map dashboard designed for tactical traffic police deployment and hot-spot mitigation.



Curating a Robust Dataset for Traffic Analysis



Transforming raw police records into a polished modeling panel

333

Raw Records

Bengaluru Traffic Police station-year records spanning 2018 to 2024.



Preprocessing

- Dropped columns with missing sub-divisions.
- Fixed critical spelling errors (e.g., White field - Whitefield).
- Purged duplicate zonal summary rows.



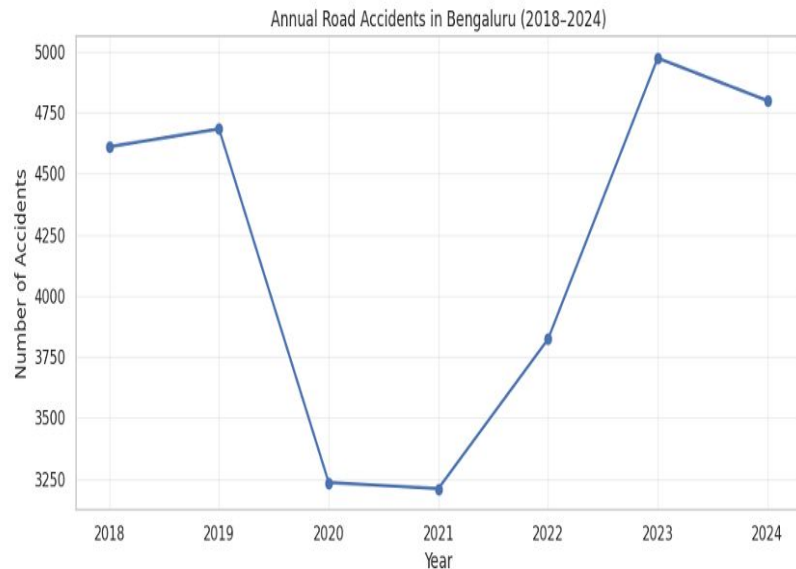
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Modelling Panel

Samples across **64 canonical** police stations, yielding **318 clean** records.

Identifying Bengaluru Crash Trends & Chronic Hotspots

Analysis of citywide patterns and 100% repeat accident zones



Citywide crash trends reveal a significant dip during the COVID pandemic between 2020 and 2021, followed by a sharp surge in 2023 and 2024, reaching approximately 14,900 crashes per year.

Analysis identified chronic 100% repeat hotspots:

1,329
TOTAL CRASHES

Kamakshipalya
Highest frequency accident cluster

1,201
TOTAL CRASHES

Peenya
Industrial corridor repeat zone

1,127
TOTAL CRASHES

Electronic City
Tech sector highway hotspot

Engineering Smart Time-Aware Features

for Risk Prediction



Engineered Signals

Engineered signals include **1-year lagged fatal and non-fatal counts**, **fatality rates**, **year-over-year percentage shifts**, and **2-year rolling averages**.

Target Class

The target class is a **Future_High_Risk** binary label, where **1** indicates a top-10 hotspot for the next year and **0** indicates low risk.



75.6%

Low Risk

vs

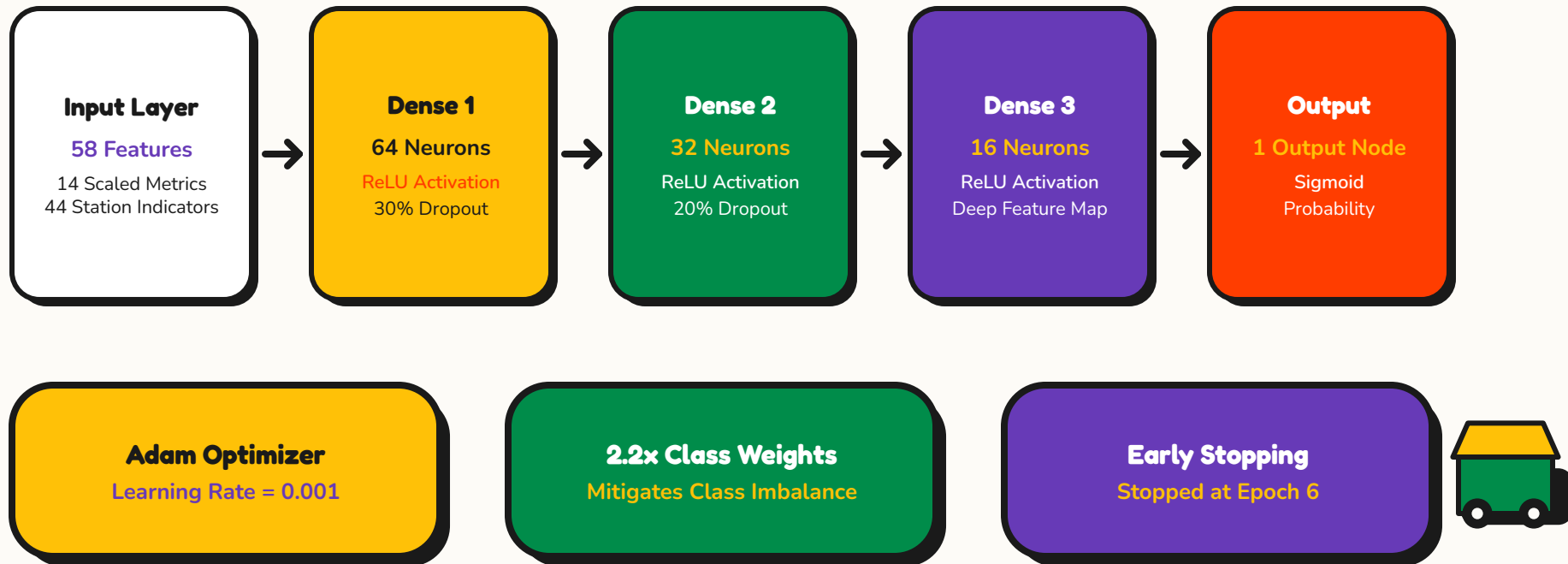
24.4%

High Risk

The dataset exhibits class imbalance, featuring 75.6% Low Risk versus 24.4% High Risk zones.

Architecting a Robust Deep Neural Network Pipeline

5-Stage Network Architecture with Class Weight Balancing



Achieving 100% Hotspot Recall on 2023 Test Set



85.29%

Accuracy

Overall test set performance

100%

Recall

Zero missed hotspots!

0.9841

ROC-AUC

Discriminative strength

0.7368

F1-Score

Robust model balance

58.33%

Precision

Low false alarms rate

22 TN

True Negatives

Safe zones identified

5 FA

False Alarms

Safe flagged as hot

0 FN

Missed Hotspots

Perfect hazard safety

7 TP

Caught Hotspots

Danger zones flagged

Zero Missed Hotspots!

Every single dangerous zone in 2023 was
successfully flagged in advance!



Ensuring Training Stability and Generalization ✨

The model demonstrates strong generalization across splits with zero-compromise safety.

86.4%

Train Accuracy

88.2%

Val Accuracy

85.3%

Test Accuracy

100%

Perfect Recall

The strategic use of **class-weighting** and **early stopping** effectively prevented overfitting on the small tabular dataset.

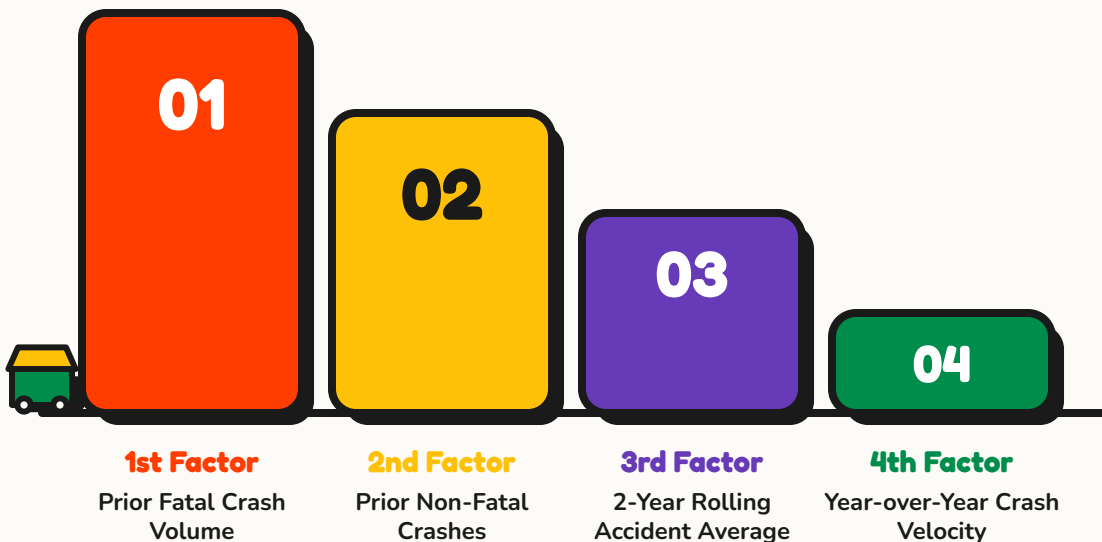
This approach ensured highly stable training and resulted in **bulletproof safety performance** for predicting road hazards!



Identifying the Core Drivers of Road Risk

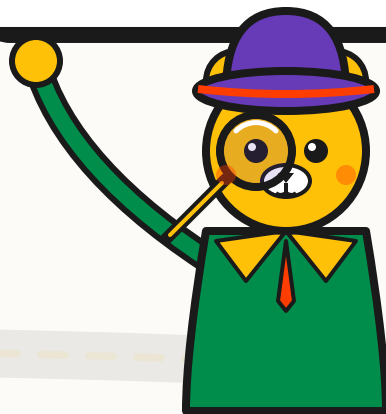


Bengaluru risk is driven by dynamic crash velocity and severity trajectory, rather than static location



Inspector's Note

Bengaluru's road risk is driven by **dynamic crash velocity** and **severity trajectory**, rather than static location identity.



Interactive Map Deployment & Risk Tiering



Live Smartphone & Dashboard GIS Interface across Bengaluru Jurisdictions

RED HIGH RISK

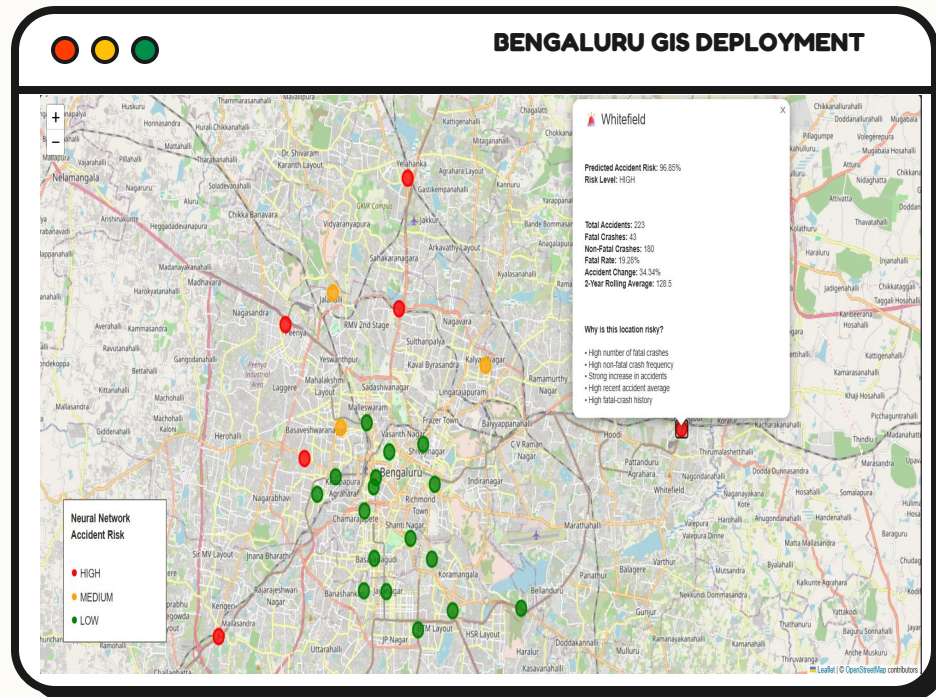
Probabilities $\geq 70\%$ - Covers 8 stations

ORANGE MEDIUM RISK

Probabilities 40% - 69% - Covers 3 stations

GREEN LOW RISK

Probabilities $< 40\%$ - Covers 17 stations



Securing a Safer Future for Bengaluru Roads



Core Impact

We successfully built a **Accident risk predictor AI early warning system** with **100% recall**, designed to enable proactive traffic management and save lives.

Future Scope

Future directions include integrating live weather and rainfall data, incorporating real-time traffic sensor feeds, and expanding to continuous crash count prediction.

Ultimately, we aim to deploy this interactive dashboard directly for the **Bengaluru Traffic Police**.

